

openedx

VPC

```
gcloud compute networks create openedx-vpc \  
  --subnet-mode=custom \  
  --bgp-routing-mode=regional
```

private subnet

```
gcloud compute networks subnets create openedx-private-subnet \  
  --network=openedx-vpc \  
  --region=asia-south1 \  
  --range=10.10.0.0/16 \  
  --enable-private-ip-google-access \  
  --secondary-range=pods-range=10.30.0.0/16,services-range=10.40.0.0/20  
e networks subnets create openedx-private-subnet --network=openedx-vpc --  
region=asia-south1 --range=10.10.0.0/16 --enable-private-ip-google-access --  
secondary-range=pods-range=10.30.0.0/16,services-range=10.40.0.0/20
```

public subnet

```
gcloud compute networks subnets create openedx-public-subnet \  
  --network=openedx-vpc \  
  --region=asia-south1 \  
  --range=10.20.0.0/16
```

firewall

```
gcloud compute firewall-rules create allow-internal \  
  --network=openedx-vpc \  
  --allow=tcp,udp,icmp \  
  --source-ranges=10.10.0.0/16,10.20.0.0/16
```

NAT

```
gcloud compute addresses create openedx-nat-ip \  
  --region=asia-south1 \  
  --network-tier=PREMIUM \  
  --project=genai-pilot-37479
```

NAT gateway

```
gcloud compute routers create openedx-nat-router \  
  --network=openedx-vpc \  
  --region=asia-south1 \  
  --project=genai-pilot-37479
```

attach cloud nat

```
gcloud compute routers nats create openedx-cloud-nat \  
  --router=openedx-nat-router \  
  --region=asia-south1 \  
  --nat-external-ip-pool=openedx-nat-ip \  
  --nat-all-subnet-ip-ranges \  
  --no-enable-endpoint-independent-mapping \  
  --project=genai-pilot-37479
```

verify

```
gcloud compute routers nats describe openedx-cloud-nat \  
  --router=openedx-nat-router \  
  --region=asia-south1 \  
  --project=genai-pilot-37479
```

get NAT

```
gcloud compute addresses list --filter="name=openedx-nat-ip"
```

Cluster

```
gcloud container clusters create openedx-cluster \  
  --region=asia-south1 \  
  --node-locations=asia-south1-c \  
  --release-channel=regular \  
  --enable-ip-alias \  
  --network=openedx-vpc \  
  --subnetwork=openedx-private-subnet \  
  --cluster-secondary-range-name=pods-range \  
  --services-secondary-range-name=services-range \  
  --enable-private-nodes \  
  --enable-private-endpoint \  
  --enable-master-authorized-networks \  
  --enable-shielded-nodes \  
  --enable-autoscaling \  
  --num-nodes=1 \  
  --machine-type=e2-standard-2 \  
  --logging=SYSTEM,WORKLOAD \  
  --monitoring=SYSTEM,POD,DEPLOYMENT,STATEFULSET,DAEMONSET,HPA,CADVISOR,KUBELET
```

```
--enable-managed-prometheus \  
--security-posture=standard \  
--enable-identity-service \  
--image-type=COS_CONTAINERD \  
--cluster-version=1.32.6-gke.1013000 \  
--workload-pool=genai-pilot-37479.svc.id.goog \  
--no-enable-legacy-authorization \  
--no-enable-basic-auth \  
--no-issue-client-certificate \  
--no-enable-l4-ilb-subsetting \  
--project=genai-pilot-37479
```

Bastion

```
gcloud compute instances create openedx-bastion \  
  --zone=asia-south1-b \  
  --machine-type=e2-medium \  
  --subnet=openedx-private-subnet \  
  --no-address \  
  --boot-disk-size=10GB \  
  --image-family=debian-12 \  
  --image-project=debian-cloud \  
  --scopes=https://www.googleapis.com/auth/cloud-platform \  
  --tags=bastion \  
  --project=genai-pilot-37479
```

```
gcloud compute firewall-rules create allow-iap-ssh \  
  --network=openedx-vpc \  
  --direction=INGRESS \  
  --action=ALLOW \  
  --rules=tcp:22 \  
  --source-ranges=35.235.240.0/20 \  
  --target-tags=bastion \  
  --priority=1000 \  
  --project=genai-pilot-37479
```